

NIAGARA INTAKE, NY, USA

WMO: 998235

Lat: 43.0770N

Lon: 79.0140W

Elev: 179

StdP: 99.19

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -15.5	(c) -13.1	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.0	(c) 28.9	(d) N/A	(e) 27.7	(f) N/A	(g) 26.6	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) N/A	(b) N/A	(c) N/A	(d) DB	(e) -17.4	(f) 31.5	(g) 2.9	(h) 1.2	(i) -19.5	(j) 32.4	(k) -21.2	(l) 33.1	(m) -22.8	(n) 33.7	(o) -25.0	(p) 34.6	(q) N/A
(4) N/A	(5) N/A	(6) N/A	(7) WB	(8) N/A	(9) N/A	(10) N/A	(11) N/A	(12) N/A	(13) N/A	(14) N/A	(15) N/A	(16) N/A	(17) N/A	(18) N/A	(19) N/A	(20) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	-4.2	-3.4	1.2	7.6	14.6	19.1	22.4	21.7	18.1	11.8	5.3	0.6
	DBStd	10.18	5.54	5.31	5.31	3.99	4.20	2.98	2.69	2.27	3.57	3.83	4.62	4.77
	HDD10.0	1666	440	374	278	93	9	0	0	0	0	25	152	294
	HDD18.3	3512	698	607	532	323	128	27	2	46	206	389	551	
	CDD10.0	1533	1	1	4	21	153	272	385	361	243	79	12	1
	CDD18.3	337	0	0	0	0	13	49	128	105	39	2	0	0
	CDH23.3	1614	0	0	1	2	75	188	739	435	170	5	0	0
	CDH26.7	229	0	0	0	0	4	17	138	49	21	0	0	0
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(15) Precipitation	PrecAvg	853	68	55	60	77	75	75	79	76	83	78	69	71
	PrecMax	1105	116	120	106	130	160	178	188	140	190	174	122	156
	PrecMin	682	37	24	19	34	20	20	18	32	38	28	27	29
	PrecStd	121	24	22	25	28	38	39	41	27	35	36	23	25
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.5	12.7	20.5	23.1	27.1	28.6	31.0	29.8	28.8	24.0	18.5	15.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	8.8	8.9	14.3	19.6	25.5	26.9	29.7	28.0	27.1	21.2	16.2	11.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	6.1	5.7	11.5	16.9	23.8	25.5	28.3	26.8	25.3	19.1	14.1	9.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	3.5	3.3	8.9	14.6	21.9	24.1	26.9	25.7	23.9	17.5	12.4	7.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	6.0	7.1	8.6	9.1	7.6	7.0	6.6	7.2	6.3	6.1	5.3
		MCDBR	8.9	9.6	11.1	13.0	11.3	9.2	8.3	7.6	8.2	8.7	8.6	7.4
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.297	0.324	0.334	0.374	0.404	0.434	0.433	0.421	0.380	0.361	0.314	0.298
	taud		2.414	2.331	2.404	2.329	2.289	2.239	2.258	2.281	2.374	2.409	2.507	2.468
	Ebn at Noon		840	868	908	891	870	842	837	836	846	816	817	806
	Edh at Noon		76	98	104	121	129	137	133	126	106	91	70	67
(44) All-Sky Solar Radiation	RadAvg		1.32	2.10	3.30	4.42	5.42	5.84	5.89	5.22	4.16	2.51	1.56	1.06
	RadStd		0.12	0.20	0.36	0.47	0.57	0.44	0.45	0.28	0.37	0.23	0.21	0.13

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page